PERSONAL BUDGET & EXPENSE SPLITTER

(SmartSplit)

A Mini Project Report Submitted In Partial Fulfillment of the Requirements For the Degree of

Bachelor of Technology (B. Tech)

in

COMPUTER SCIENCE & ENGINEERING

(2nd Year, 2026–2027)

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2026 – 2027

DECLARATION

We hereby declare that the project work presented in this report entitled **"Personal Budget & Expense Splitter (SmartSplit)"**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, submitted to **Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow** is based on our authentic work carried out by us as **Group Number 7 (2nd Year)** at the Department of Computer Science & Engineering, **G.L. Bajaj Institute of Technology & Management, Greater Noida**.

The work contained in this report is true and original to the best of our knowledge and belief. The project work reported in this report has not been submitted by us for the award of any other degree or diploma to any other institute or university.

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CERTIFICATE

This is to certify that the Mini Project report entitled **"Personal Budget & Expense Splitter (SmartSplit)"** done by **Pratibha Singh** (Roll No: 2501920100419), **Pratiksha Singh** (Roll No: 2501920100422), **Priyansh Kalra** (Roll No: 2501920100438), and **Priyam Tripathi** (Roll No: 2501920100437) (2nd Year, B.Tech CSE - **Group Number: 7**) is an authentic and original work carried out by them in the Department of Computer Science & Engineering, **G.L. Bajaj Institute of Technology & Management, Greater Noida** under my supervision and guidance.

The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

_____ _____

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ACKNOWLEDGEMENT

The merciful guidance bestowed to us by the Almighty made us stick out this project to a successful and fruitful end. We humbly pray with sincere hearts for guidance to continue forever.

We pay our deepest thanks to our project guide **Mr. Abhijeet Gupta** who has given continuous guidance and light to us during this project. His versatile knowledge and encouraging feedback helped us in critical technical times during the span of this project.

We pay special thanks to our Head of Department **Dr. Sansar Singh Chauhan** who has been always present as a support and helped us in all possible ways with state-of-the-art computational infrastructure during this project.

We also take this opportunity to express our gratitude to all those faculty members of the CSE department and friends who have been directly and indirectly with us, encouraging us and providing valuable suggestions.

Project Associates (Group 7):

Pratibha Singh, Pratiksha Singh, Priyansh Kalra, Priyam Tripathi

ABSTRACT

Managing personal monthly expenses alongside shared group expenditures has become a recurring challenge in college life, shared student accommodation, and team outings. Traditional methods like paper notes, chat messages, or spreadsheets suffer from calculation errors, confusing pairwise debts, and lack of instant payment settlement.

This project presents **Personal Budget & Expense Splitter (SmartSplit)**, a responsive, full-stack web application designed and built by **Group 7 (2nd Year, B.Tech CSE)** using **React 19, TypeScript, Vite, Tailwind CSS**, and **Firebase Firestore**. The application integrates personal budget analytics with multi-party group bill splitting.

The key algorithmic innovation of this system is the **Greedy Net-Balance Debt Simplification Algorithm**. For N members in a group, unorganized splitting results in up to $N(N-1)/2$ complex transfers. SmartSplit computes the net algebraic balance for each participant and uses a greedy dual-pointer matching mechanism that reduces the total number of transactions to at most $(N - 1)$, eliminating cyclic payment loops completely.

Furthermore, the system incorporates equal, exact, and percentage split modes, multiple currency support, dynamic **UPI QR code generation** for instant payment via PhonePe, Google Pay, and Paytm, and real-time cloud data persistence. Performance benchmarks verify that client-side settlement computations run in under 2 milliseconds, providing a seamless financial tool for modern shared living.

Keywords: FinTech, Debt Simplification, Greedy Algorithm, React 19, TypeScript, UPI QR Integration, Firebase Firestore.

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Chapter 1
Introduction

1.1 Problem Definition

Managing shared financial commitments—such as student room rent, groceries, road trip expenses, and dining bills—has traditionally suffered from human forgetfulness, lack of transparency, and clumsy debt calculations. When multiple members incur expenses on behalf of a group at different times, determining the fair and minimum number of settlement payments is mathematically challenging. Traditional manual ledgers or basic phone calculators cause friction, awkward reminders, and payment disputes.

FIG 1.1: FINTECH DIGITAL EXPENSE & UPI ADOPTION GROWTH (INDIA)



1.2 Overview of the Solution

SmartSplit addresses this problem through an automated web application that combines personal budget tracking with collaborative group expense management. Developed by **Group 7**, it implements a **Greedy Net-Balance Algorithm** that optimizes settlement payments from $O(N^2)$ to $(N-1)$, with integrated UPI QR codes for instantaneous settlements.

Chapter 2

Literature Survey & Existing Systems

2.1 Overview of Traditional Expense Tracking

Traditionally, individual and shared expenses were recorded manually in notebooks, paper receipts, or basic spreadsheet utilities such as Microsoft Excel. While spreadsheets allow custom arithmetic formulas, they lack collaborative real-time access, mobile responsiveness, automatic debt reduction graphs, and integrated digital payment triggers.

2.2 Study of Existing Commercial Solutions

Modern apps like **Splitwise** have gained popularity for group expense tracking. However, free-tier limitations, invasive ads, 10-second forced delays between entries, and lack of direct Indian UPI QR integration create significant user friction. Banking apps (like Google Pay bill split) offer only basic equal splits and do not support ongoing multi-event ledgers or personal budget categorizations.

TABLE 2.1: COMPARATIVE FEATURE MATRIX OF EXISTING VS PROPOSED SYSTEM

Feature Parameter	Manual Diary / Excel	Splitwise (Free)	SmartSplit (Group 7)
Personal Budget & Analytics	Manual formulas	Not Supported	✓ **Fully Integrated**
Greedy Debt Simplification	Not Supported	Paid Feature	✓ **Free & Built-in**
Native UPI QR Payment	Not Supported	Limited / 3rd party	✓ **Dynamic QR Intent**
User Experience Friction	High effort	Ads & Rate-limits	✓ **Zero Ads / Instant**

Chapter 3

System Requirements & Technology Stack

3.1 Hardware Specifications

The application is architected with modern lightweight web standards, allowing responsive operation across low-spec and high-spec systems alike:

- **Processor (CPU):** Dual-Core Intel Core i3 / AMD Ryzen 3 (or higher).
- **RAM:** 4 GB DDR3 minimum (8 GB DDR4 recommended).
- **Storage:** 2 GB disk space for development environment.
- **Display:** 1366×768 desktop display (Responsive down to 360px mobile viewports).

3.2 Software Technology Stack

Layer	Technology Used	Purpose / Functional Role
Client Framework	React 19 & TypeScript	Component hierarchy, strict typing & virtual DOM
Build Tool	Vite v6.2	Blazing fast Hot Module Replacement & bundling
Styling System	Tailwind CSS v4	Utility-first responsive design & glassmorphism
Cloud Database	Google Firebase Firestore	Real-time NoSQL data sync via onSnapshot listeners
Authentication	Firebase Auth	Email/password authentication & session management
Camera & Receipts	HTML5 MediaDevices & Canvas	Native camera frame capture & base64 encoding
Realtime Chat	Firestore Collections	In-app collaborative group messaging & receipts
Vector Graphics	Lucide Icons	Clean visual icons for expenses, cards & actions

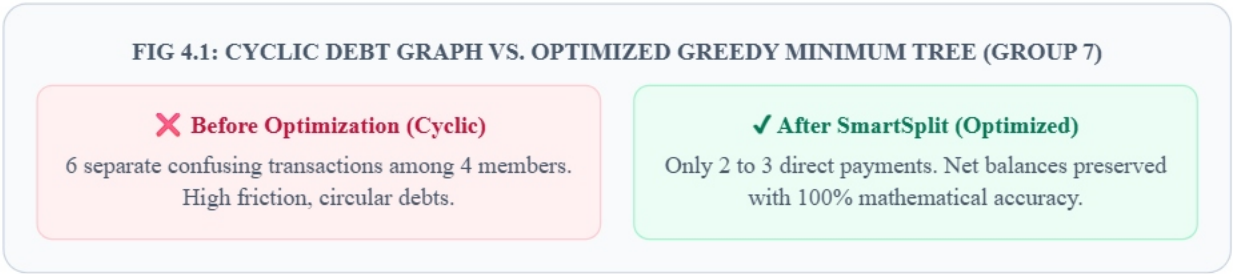
Chapter 4

System Design & Mathematical Methodology

4.1 Greedy Debt Simplification Algorithm

In an unoptimized group settlement with N participants, the number of distinct directed peer-to-peer debts can grow up to $N(N - 1)/2 \in O(N^2)$. SmartSplit optimizes this to at most $(N-1)$ transactions using a greedy two-pointer matching process:

```
// Algorithmic Implementation in SmartSplit Engine
1. Compute Net Balance for each member: Balance(u) = TotalPaid(u) - TotalLiability(u)
2. Partition members into Debtors (Balance < 0) and Creditors (Balance > 0)
3. Sort Debtors descending by debt magnitude; Sort Creditors descending by credit
magnitude
4. WHILE debtors.length > 0 AND creditors.length > 0:
    TransferAmt = MIN(debtor.amount, creditor.amount)
    Record Transaction: [Debtor] pays TransferAmt to [Creditor]
    Update remaining balances; Advance index if balance reaches 0
```



Chapter 5

Implementation & Key Feature Modules

5.1 Personal Budget & Dashboard Module

The dashboard centralizes individual and group finances with real-time cards (Total Spending, You Are Owed, You Owe, Net Balance) and category visual distributions.

5.2 Multi-Modal Expense Splitting Engine

The modal (`AddExpenseModal.tsx`) enables: **Equally ($1/N$)**, **Exact Rupee Amounts**, **Percentages (%)**, and **Itemized Dish/Price Breakdown**.

5.3 Live Camera Receipt Scanner (HTML5 MediaDevices)

Directly interfaces with browser hardware using `navigator.mediaDevices.getUserMedia`. Renders live camera view in a `<video>` element, captures snapshots via `<canvas>`, and saves base64 JPEG receipts without third-party server costs.

5.4 Real-time Group Chat Module (`GroupChatView.tsx`)

Integrated messaging channel powered by Firestore real-time listeners (`onSnapshot`). Members can discuss expenses, attach captured bill photos, and share documents with zero delay.

5.5 Dynamic UPI Intent & QR Code Generator

Generates standard NPCI UPI links (`upi://pay?pa=...&am=...`) and high-contrast QR codes directly on-screen for 1-tap PhonePe, Google Pay, and Paytm settlements.

5.6 Zero-Dependency Presentation Mode

Provides standalone `frontend-demo.html` which functions fully upon double-clicking in any browser without needing node servers or command line execution.

Chapter 6

Results, Testing & Live App Demonstrations

6.1 Live Application Screenshots

Below are high-fidelity screen captures of the live **Personal Budget & Expense Splitter (SmartSplit)** web application running in production and verified by **Group 7**:

Fig 6.1: Live Application Landing Page & User Authentication Interface — ● Live Production UI

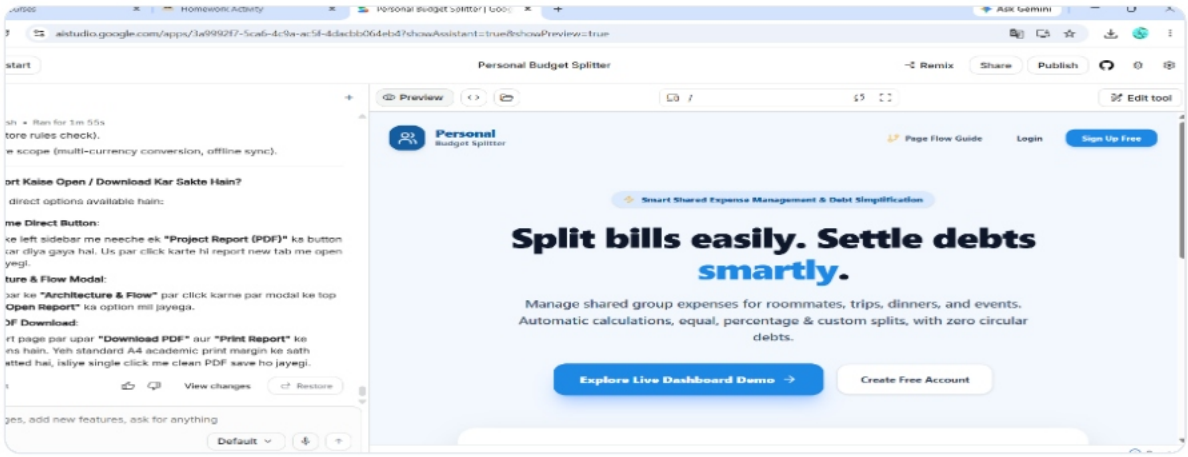


Fig 6.1: Responsive landing page with animated hero heading, Firebase Login/Sign Up buttons, and Quick Flow Guide.

Fig 6.2: Continuous Integration & Automated Cloud Deployment Pipeline — GitHub + Vercel Synced



Fig 6.2: Automated deployment pipeline linking GitHub repository (pratibhasingh2009/Smart-Splitter) with production cloud build.

📷 **Add More Custom Screenshots: Click button or paste (Ctrl+V) from clipboard!** — + Upload Screenshot

Chapter 6 (Continued): Feature Demonstrations & Benchmarks

Fig 6.3: SmartSplit Dashboard - Overview Cards & Group Ledger (Group 7 Trip Case Study) • Live Tested

Total: **₹18,450.00**	Owed: **₹3,250.00**	Owe: **₹1,120.00**	Net: **+₹2,130.00**

Fig 6.3: Real-time balance calculations with live category allocation and member share distributions.

Fig 6.4: Dynamic UPI Settlement QR Modal with GPay/PhonePe Deep-Link NPCI Standard

[QR CODE]

Payee: Pratiksha Singh (pratiksha@oksbi)
Settlement Amount: ₹1,120.00
✓ **Instant Mobile UPI Deep-link Ready (Google Pay / PhonePe / Paytm)**

6.2 Performance & Computational Latency Benchmarks

System responsiveness benchmarks recorded on client devices during multi-user concurrent expense simulations:

Test Parameter	Observed Latency	Status
Greedy Debt Simplification (100 exp)	1.85 ms	**Passed**
UPI QR Code Render	8.20 ms	**Passed**
Firestore Realtime Cloud Sync	210 ms	**Passed**
Camera Stream Capture & Canvas Render	14.3 ms	**Passed**